

PERSONAL INFORMATION



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Address: TPR2 05.22, 163 Av. de Luminy, 13009 Marseille.
Personal website: <https://xys220.github.io/>
Team: Past memebre of CANA (Natural Computation) - Laboratoire d'Informatique et Systèmes (LIS UMR 7020).

EDUCATION

12/2022 -05/2026

Doctoral (Doctorat en Informatique). Département de Mathématiques et Informatique de Marseille, Aix-Marseille Université, FR.

Thesis: “*Quantum perceptron algorithms and machine learning techniques for quantum separability*”. Defended on 27th May 2026.

Thesis directors: [Giuseppe Di Molfetta](#) and [Hachem Kadri](#).

10/2020 -10/2021

Master of Science (Quantum Fields and Fundamental Forces -Merit). Department of Physics, Imperial College London, UK.

MSc dissertation: “*Topological Magnetic Monopoles and Electroweak Sphalerons*”.

Dissertation supervisor: [Arttu Rajantie](#).

09/2016 -06/2020

Bachelor of Science (Materials Physics -Class One Honours). Department of Physics and Astronomy, the University of Sheffield, UK.

Bachelor of Engineering (Materials Science and Engineering -Average Mark: 92.27/100). Department of Overseas Education, Nanjing Tech University, China.

PUBLICATIONS

1. Tian, J., Sun, X., Du, Y., Zhao, S., Liu, Q., Zhang, K., ... & Tao, D. (2023). *Recent advances for quantum neural networks in generative learning*. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(10), 12321-12340.
2. Sun, X., Roget, M., Di Molfetta, G. et al. *On quantum perceptron learning via quantum search*. Quantum Mach. Intell. 8, 68 (2026). <https://doi.org/10.1007/s42484-026-00406-4>.
3. *Large-Scale Quantum Separability Through a Reproducible Machine Learning Lens*. Balthazar Casalé*, Xiaoyu Sun*, Giuseppe Di Molfetta, Sandrine Anthoine, Hachem Kadri. (* for equal contribution, submitted to Quantum, August 2026). Codes available on <https://gitlab.lis-lab.fr/hachem.kadri/ML-Quant-Sep>.

PARTICIPATION IN MEETINGS, SCHOOLS AND WORKSHOPS

- ❖ Ecole d'été d'informatique (2023 summer school <https://ecoles-cea-edf-inria.fr/history/ecole-informatique-2023/>).
- ❖ Quantum days at AMU (workshop 2023 <https://conferences.cirm-math.fr/3130.html>).
- ❖ Mathematics and Physics of Quantum Computing and Quantum Learning (mpqcql 2025 summer school <https://mpqcql2025.sciencesconf.org/>): presented a poster.

- ❖ Quantum days at AMU (workshop 2025 <https://conferences.cirm-math.fr/3782.html>): presented a short talk.
- ❖ Journées Informatique Quantique (JIQ 2026 <https://quantique.labri.fr/jiq2026/>): presented a talk ([slides](#)).

INTERNSHIP & RESEARCH EXPERIENCE

10/2022- 06/2022	Interned at Yanqi Lake Beijing Institute of Mathematical Sciences and Applications (BIMSA) with Zhengwei Liu. ➤ Researched quantum information and quantum error correction codes.
06/2022- 12/2021	Interned at the quantum group at JD explore Academy, Beijing, China, worked on quantum machine learning (specifically quantum generative model) with Yuxuan Du and Xingyao Wu. ➤ Published a survey: “<i>Recent Advances for Quantum Neural Networks in Generative Learning</i>” with co-authors. ➤ Researched adversarial attacks in quantum generative learning models.
10/2017- 11/2018	Served as student team leader on a undergraduate research project “Undergraduate Training Program for Innovation and Entrepreneurship of Jiangsu Province”, with the project “ <i>Research on High Performance LED Based on Multi - Quantum Well Perovskite</i> ”.
09/2017	Studied at the Nanomaterials Laboratory of Nanjing Tech University as a member.

LANGUAGES

Chinese	Native language
English	IELTS (year 2020)- Listening: 7.0 Reading: 8.5 Writing: 6.0 Speaking: 6.0 Overall: 7.0 (C1 equivalent)
French	A2

PROGRAMMING SKILLS

Certificates	Jiangsu Computer Rank Exam Certificate (Jiangsu Province, China) - Grade I. National Computer Rank Exam Certificate (China) - Grade II.
Coding	python, C.